Journal of Physics B: Atomic,
Molecular and Optical Physics

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To cite this article: F X Gadea and A Boutalib 1993 *J. Phys. B: At. Mol. Opt. Phys.* **26** 61

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J. Phys. B: At. Mol. Opt. Phys. 26 (1993) 61-74. Printed in the UK

Computation and assignment of radial couplings using accurate diabatic data for the LiH molecule

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Received 1 June 1992, in final form 24 September 1992

Abstract. The radial couplings between the $^{3,1}\Sigma^+$ adiabatic states dissociating into Li(2s, 2p, 3s, 3p, 3d, 4s and 4p) + H and $Li^+ + H^-$ are determined from accurate diabatic data. The results are validated by a comparison with previous *ab initio* calculations. It is shown that the peaks of the radial coupling can be interpreted from the diabatic curves. An over-repulsiveness of the valence diabatic states as compared to the Rydberg diabatic states generates, at short internuclear distances, a series of crossings between neutral diabatic states which are associated with intense peaks of the radial coupling both for the singlet and triplet Σ states.

1. Introduction

The Born-Oppenheimer approximation plays a central role in molecular calculations; however there are a large variety of problems where it breaks down. This is the case when several electronic states dynamically interact, as occurs at conical intersections, or at avoided crossings. Radiationless transitions represent the signature of such phenomena and are in fact rather common in molecular dynamics. In most cases, two equivalent representations can be adopted for the theoretical treatment of such problems. These are: the adiabatic representation where the radial coupling causes the non-adiabatic transitions and the diabatic representation where the same role is imported to the electronic coupling.

Each representation is subject to well known limitations and difficulties centred on the computation of the radial coupling (when possible) for the adiabatic representation and on the production of electronic states as diabatic as possible (i.e., presenting a radial coupling as small as possible) for the diabatic representation. They also have their own advantages, mainly a unique and unambiguous definition for the adiabatic representation leading to the familiar notion of potential energy and a more transparent physical picture related to the appealing valence bond configurations, combined to numerical interests, for the diabatic representation. Both these electronic representations are currently used in dynamical calculations devoted to molecular systems.

In the adiabatic picture, the *ab initio* radial coupling can be evaluated by several methods (see Mendez *et al* [1]) which can be divided into two groups:

(i) numerical methods, mainly finite difference techniques [2-7];

(ii) analytical expressions involving the nuclear position operators [8-10].

Our present approach to determine the radial coupling between adiabatic states is different. We start from the knowledge of reliable diabatic states and assume them to be strictly diabatic. Thus we can obtain the radial coupling through a numerical differentiation of the rotation matrix or through the Hellmann-Feynman expression.

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Both methods here give the same coupling since the adiabatic states result from the exact diagonalization of the small matrix of the electronic Hamiltonian operator in the diabatic basis set. Of course the reliability of this approach is completely conditioned by the diabaticity quality of the diabatic states, since we definitely neglect the residual coupling between diabatic states. On the other hand, the obtention of reliable radial couplings allows a direct confirmation of the validity of the diabatization method used. An interesting aspect is that this approach allows us to correlate the main peaks of the radial coupling with energy crossings between the diabatic states, giving a physical insight to this radial coupling responsible for the radiationless transitions. We found that two kinds of crossings contribute to the main features of the radial couplings: the well known neutral–ionic crossings usual in the alkali hydride molecules and neutral–neutral crossings which come out from this study and were beyond suspicion from the adiabatic data.

2. Results and discussion

The diabatization method [11], the results of which are used here, rests upon effective Hamiltonian theory combined to an effective overlap matrix. Effective Hamiltonian theory is used in an unusual way [6, 12, 13] where the diabatic states lie in the target space and are linear combinations of the adiabatic states, allowing for variational properties of the effective Hamiltonian operator [14]. An effective overlap matrix is built between the set of atomic orbitals generating the reference states which span the model space and the set generating the adiabatic states which span the target space. For the CsH molecule, this method was proved to produce vanishingly small residual couplings between the diabatic states resulting from a limited CI expansion [11]. The accurate adiabatic and diabatic results we used here have been presented elsewhere [15]. They are reproduced in figures 1–4; $^1\Sigma^+$ diabatic, $^1\Sigma^+$ adiabatic, $^3\Sigma^+$ diabatic and $^3\Sigma^+$ adiabatic respectively. The adiabatic energies satisfactorily compared to previous *ab initio* calculations and the resulting spectroscopic constants and vibrational levels reproduce the experimental data with an excellent overall accuracy (less than 1% error).

We have evaluated the radial coupling according to two methods. Denoting the adiabatic states by $|\psi_i\rangle$ (with associated energy E_i), the diabatic states by $|\phi_k\rangle$ and the unitary matrix connecting the two sets by $\mathbf{C}$ ($\mathbf{C}_{ki} = \langle\phi_k|\psi_i\rangle$)

$$|\psi_i\rangle = \sum_k \mathbf{C}_{ki}|\phi_k\rangle.$$

(i) Numerical differentiation of the rotation matrix

$$\langle\psi_i|\frac{\partial}{\partial R}|\psi_j\rangle = \sum_\alpha \mathbf{C}_{i\alpha}\frac{\partial \mathbf{C}_{\alpha j}}{\partial R}$$

or in matrix notation

$$= \left\langle \mathbf{C}^+ \frac{\partial \mathbf{C}}{\partial R} \right\rangle_{ij}.$$

(ii) Hellmann–Feynman expression

$$\langle\psi_i|\frac{\partial}{\partial R}|\psi_j\rangle = (E_j - E_i)^{-1}\langle\psi_j|\frac{\partial \mathbf{H}}{\partial R}|\psi_i\rangle$$

$$= (E_j - E_i)^{-1}\left\langle \mathbf{C}^+ \frac{\partial \mathbf{H}}{\partial R}\mathbf{C} \right\rangle_{ij}$$

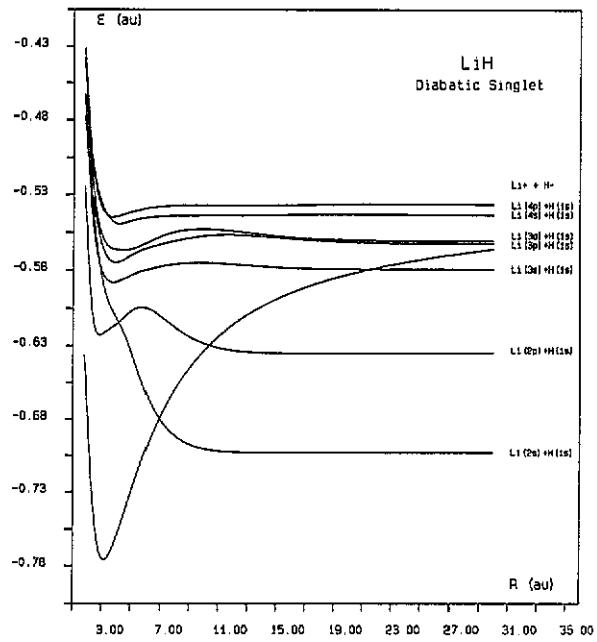


Figure 1. $^1\Sigma^+$ diabatic curves of the LiH molecule (from [15]).

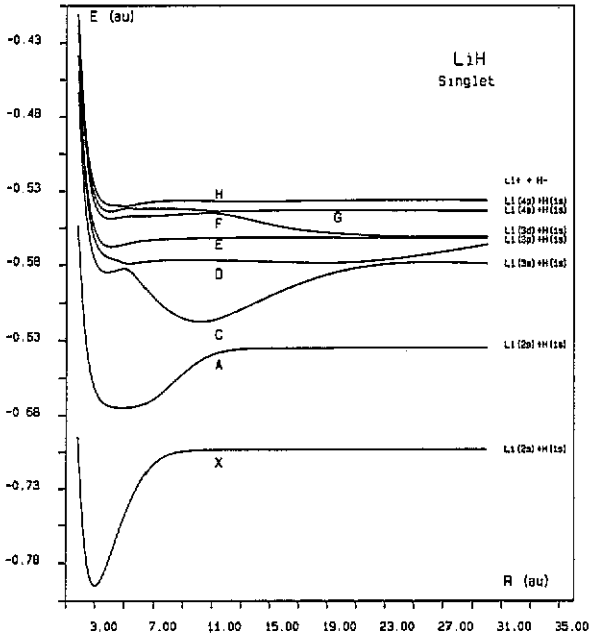


Figure 2. $^1\Sigma^+$ adiabatic curves (from [15]).

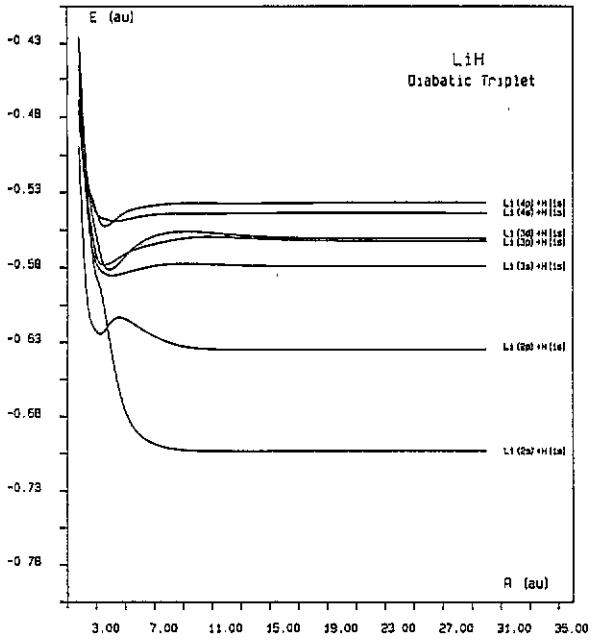


Figure 3. $^3\Sigma^+$ diabatic curves (from [15]).

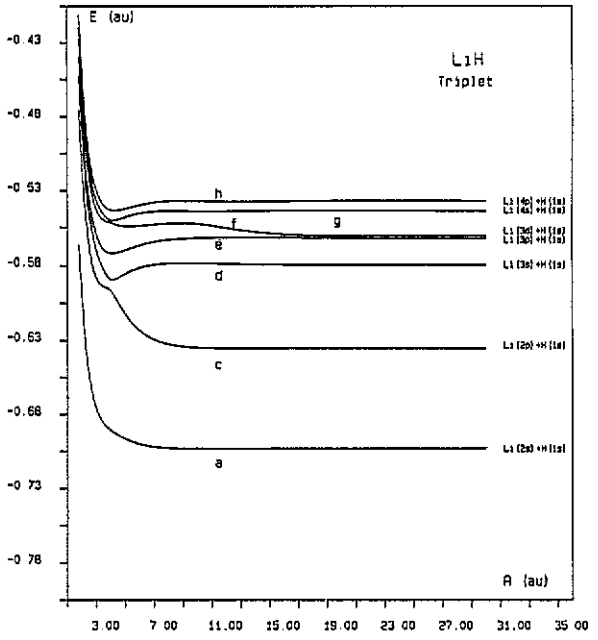


Figure 4. $^3\Sigma^+$ adiabatic curves (from [15]).

where the electronic Hamiltonian is assumed to be known in and restricted to the diabatic basis, i.e.

$$\mathbf{H}_{kl} = \langle \phi_k | \mathbf{H}_{el} | \phi_l \rangle$$

$$\left\langle \frac{\partial \mathbf{H}}{\partial R} \right\rangle_{kl} = \langle \phi_k | \frac{\partial \mathbf{H}_{el}}{\partial R} | \phi_l \rangle = \frac{\partial \mathbf{H}_{kl}}{\partial R}.$$

As expected, both methods here lead to the same result. In both cases the matrix elements $\mathbf{H}_{ij}$ were interpolated by cubic splines in the first step and we determined the **C** matrix by diagonalization at all distances required by the three-points numerical differentiation. Interpolation of the rotation matrix **C** leads to numerical instability due to the loss of unitarity and should be avoided.

We treated nearly all states dissociating below the ionic limit, i.e. the neutral states Li(2s, 2p, 3s, 3p, 3d, 4s and 4p)+H(1s) and the ionic one Li^+H^-. For the diabatic diagonal energy of the ionic state we tried two corrections accounting for an underestimation of the H electron affinity due to basis set limitations:

(i) a constant correction corresponding to our error for the ionic limit;

(ii) a distance-dependent correction where the basis set superposition error for H^- in presence of the lithium basis set was taken into account.

These corrections are small, however, as illustrated in figure 5; they allow for an identification of the contribution of the ionic state to the peaks of the radial coupling. In this figure there are two peaks, one at short and the other at large internuclear distances. For the inner peak, all three curves are identical, this coupling is thus not related to the ionic state, while the position of the outer peak is clearly affected by the corrections on the ionic diabatic energy, and corresponds to neutral ionic interactions. This assignation is evident from figure 1, the inner peak corresponds to the 2s–3s diabatic crossing while the outer one to the 3s–ionic crossing. However, for other neighbour and next-neighbour couplings the corrections of the ionic curve facilitate

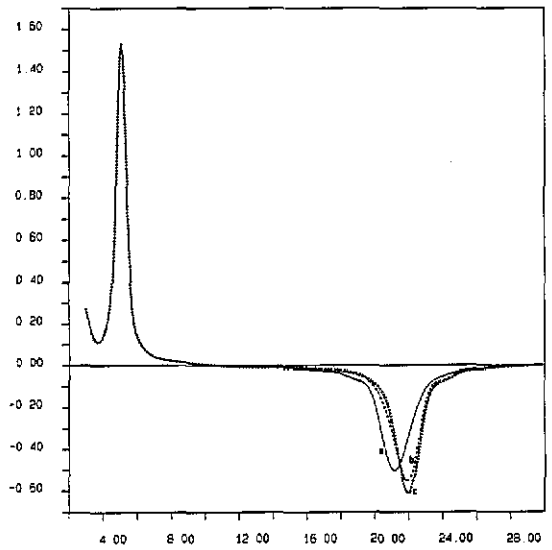


Figure 5. Radial coupling between the neighbour adiabatic $^1\Sigma^+$ states number 3–4 resulting from (a) the *ab initio* data without any correction; (b) an *R*-dependent correction of the ionic diabatic energy; (c) a constant correction of the ionic diabatic energy.

the identification of the contributions. For all other couplings only the curves corresponding to the *ab initio* data without any correction are reported.

Figures 6–11 show the radial coupling between neighbour ($\langle\psi_i|\partial/\partial R|\psi_j\rangle, |i-j|=1$) and next-neighbour ($|i-j|=2$) $^1\Sigma^+$ adiabatic states. The neighbour and next-neighbour couplings between the $^3\Sigma^+$ adiabatic states (labelled from 9 to 15) are reported in figures 12–16. The other radial couplings between $^1\Sigma^+$ and between $^3\Sigma^+$ adiabatic states are much smaller and were not reported.

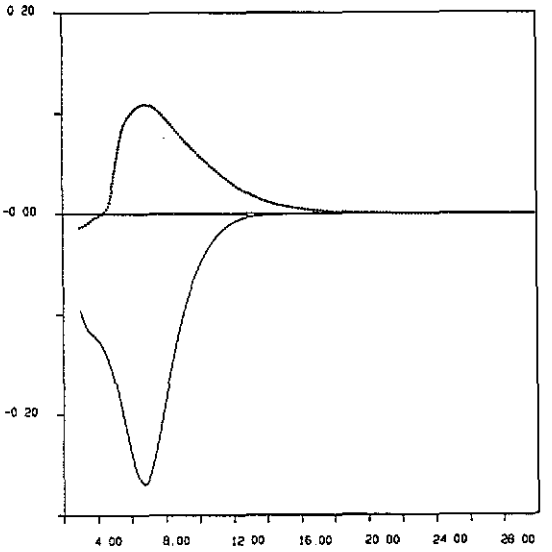


Figure 6. Radial coupling between the adiabatic $^1\Sigma^+$ neighbour states number 1–2 (full line) and the next-neighbour 1–3 ones.

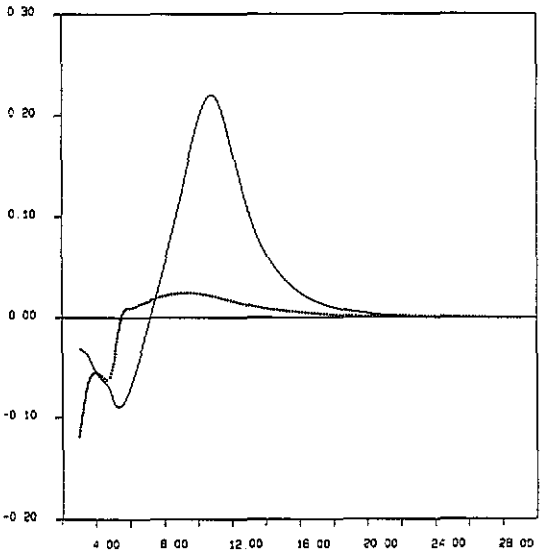


Figure 7. As figure 6 for the neighbour 2–3 and next-neighbour 2–4.

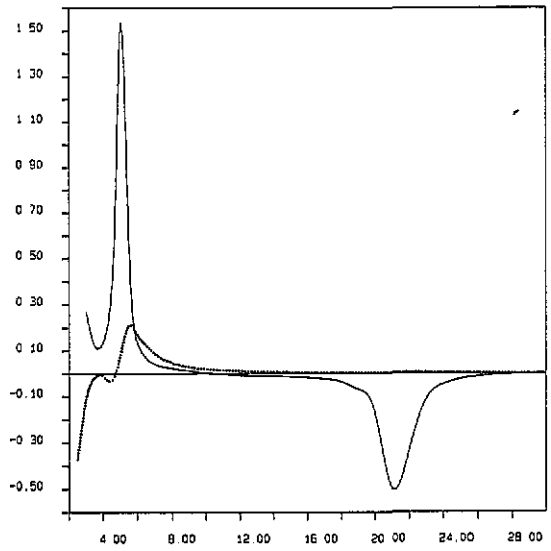


Figure 8. As figure 6 for the neighbour 3-4 and next-neighbour 3-5.

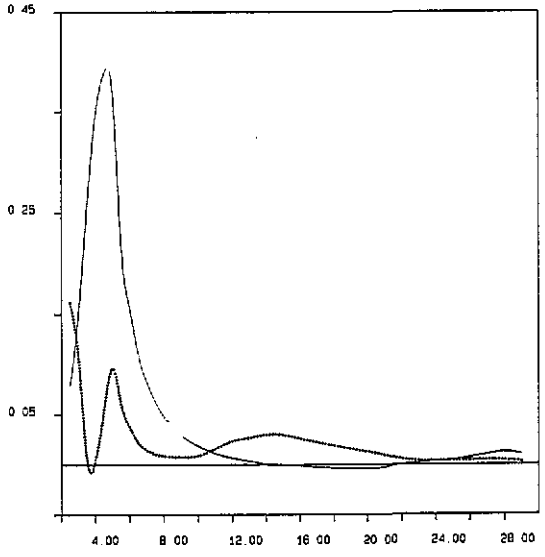


Figure 9. As figure 6 for the neighbour 4-5 and next-neighbour 4-6.

Mendez *et al* have used various techniques to compute directly the radial coupling 1-2, 2-3, 3-4 between the $^1\Sigma^+$ adiabatic electronic states ψ_1 and ψ_2, ψ_2 and ψ_3, ψ_3 and ψ_4 respectively. Our couplings 1-2, 2-3 and 3-4 are very similar in shape, location and magnitude to this completely different calculation. Since the radial coupling is known to be very sensitive to the details of the adiabatic wavefunctions, this similarity brings a confirmation for the results themselves and also a new confirmation of the validity of the diabatization procedure we used.

We can now try to learn some new things from these data, establishing a correlation between the peaks of the radial coupling and the diabatic curves. The intense peaks

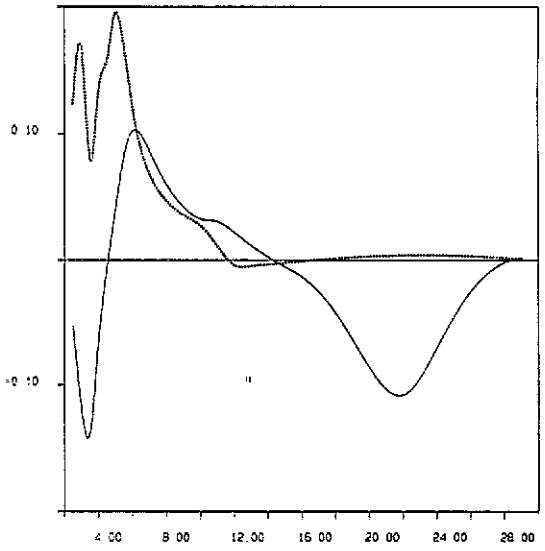


Figure 10. As figure 6 for the neighbour 5–6 and next-neighbour 5–7.

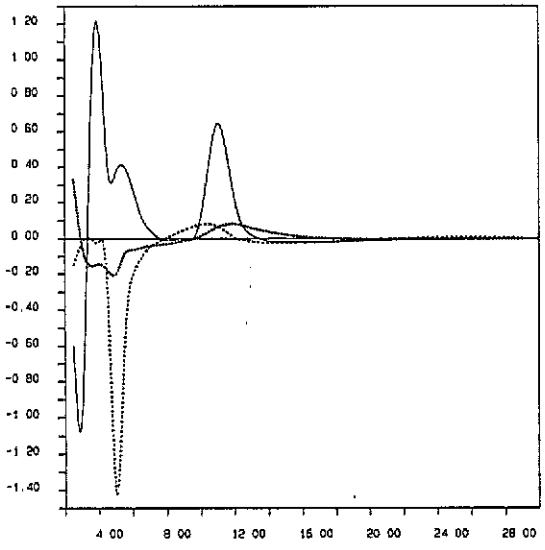


Figure 11. As figure 6 for the neighbour 6–7 (full curve) and 7–8 (chain curve), and the next-neighbour 6–8 (dotted curve).

at 7 au for the 1–2 coupling, at 11 au for the 2–3 coupling and 21 au for the 3–4 one have been assigned by Mendez *et al* [1] to neutral–ionic pseudo crossings between the ionic Li^+H^- and the neutral Li(2s, 2p, 3s)H main configurations respectively. Inspection of the diabatic curves and the sensitivity of these peaks to the corrections for the ionic curve (see figures 5–7) clearly confirm this assignment. However, the internal peaks around 5 au for both 2–3 and 3–4 couplings could not be interpreted from the adiabatic data. Inspection of the diabatic $^1\Sigma^+$ curves plotted in figure 1 reveals that they should be attributed to the crossing between the neutral states Li(2s)H and Li(2p)H for the peak of the 2–3 radial coupling and between the neutrals Li(2s)H

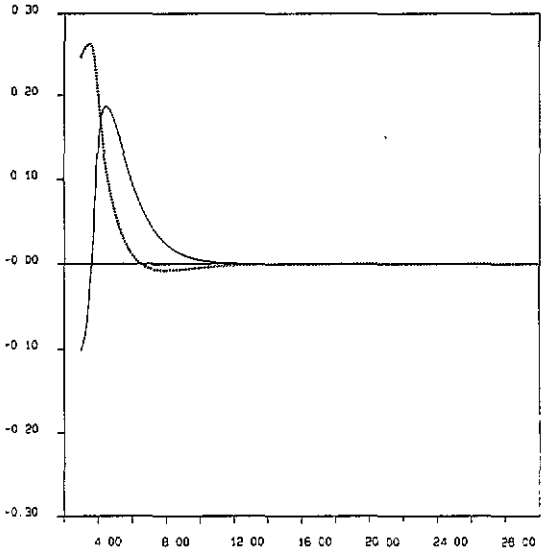


Figure 12. *Radial coupling between the lowest adiabatic* $^3\Sigma^+$ *neighbour states labelled* 9–10 and the next-neighbour ones labelled 9–11.

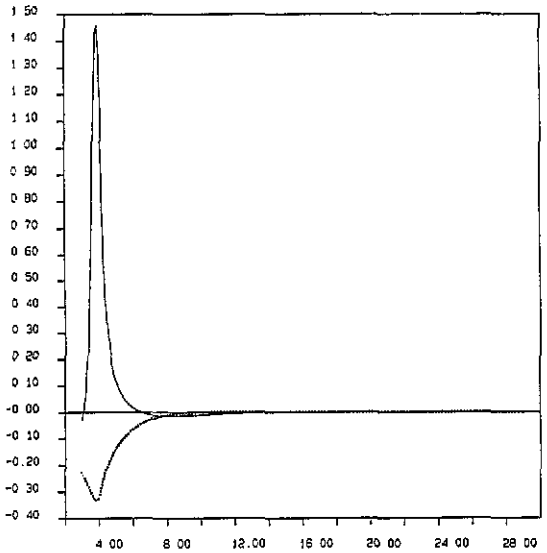


Figure 13. As figure 12 for the neighbour 10–11 and next-neighbour 10–12.

again but with Li(3s)H now for the 3–4 coupling. Our results are rather similar to those of Mendez *et al* [1] but some significant differences remain, a complete agreement would have been absolutely unexpected. The agreement is especially good for the neutral–ionic peaks and is better with their *ab initio* data than with their model potential calculation. The largest discrepancy is for the neutral–neutral crossing of the 3–4 coupling; we found it somewhat less wide and more intense, at very short distances our coupling presents a new peak while only a shoulder is apparent in the results of Mendez *et al* [1]. We cannot speculate too much about such differences since they

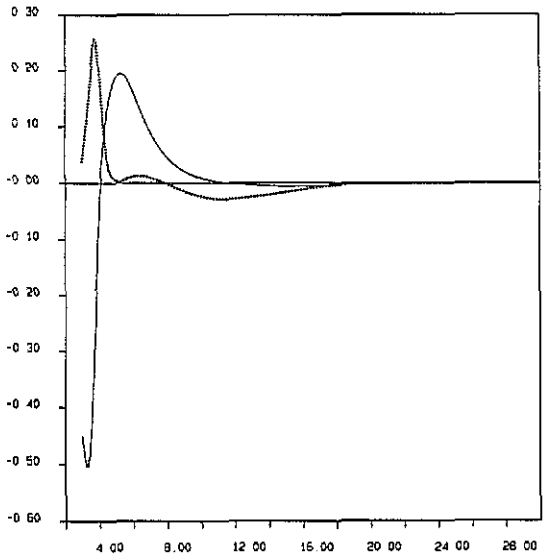


Figure 14. As figure 12 for the neighbour 11-12 and next-neighbour 11-13.

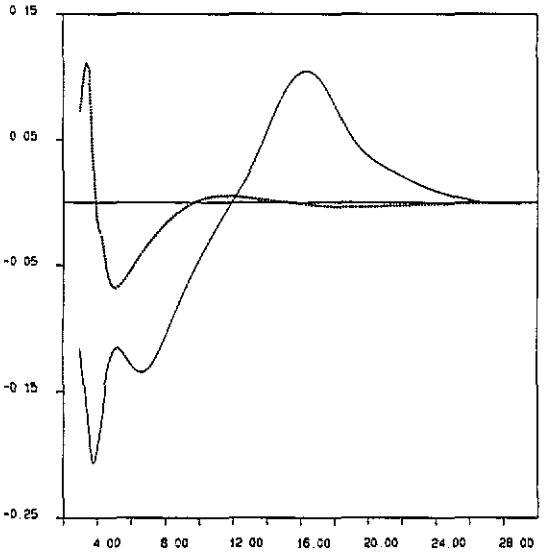


Figure 15. As figure 12 for the neighbour 12-13 and next-neighbour 12-14.

correspond to the strongly repulsive part of the curves where our *ab initio* energy calculations may suffer from some limitations, since we have not explicitly treated the lithium core but we used a pseudopotential and a core-valence correlation operatorial approach.

For both 2-3 and 3-4 couplings, the internal main peak is due to the fact that the valence diabatic $^1\Sigma^+$ Li(2s)H state is much more repulsive than the Rydberg states, leading to a cascade of neutral-neutral crossings at short internuclear distances. This behaviour is rather unperceivable in the adiabatic picture and even confusing. For

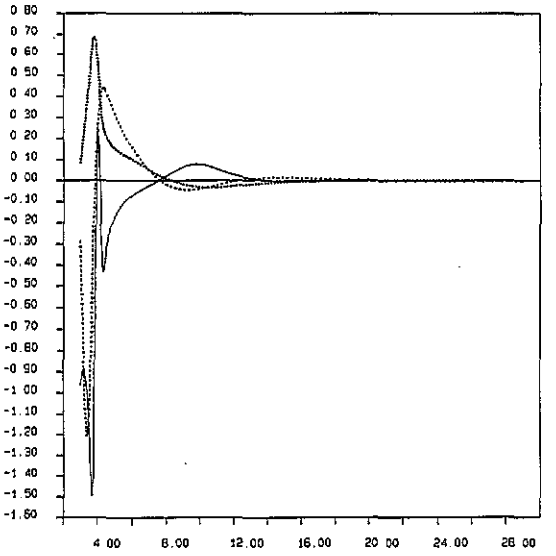


Figure 16. As figure 12 for the neighbour 13–14 (full curve) and 14–15 (chain curve) and the next-neighbour 13–15 (dotted curve).

example the internal avoided crossing between the $^1\Sigma^+$ curves labelled C and D in figure 2 (adiabatics 3–4), would have been intuitively attributed to a crossing between Li(2p)H and Li(3s)H quasidiabatic curves when considering only the adiabatic curves. Moreover the electronic coupling could have been estimated as half the smallest energy difference. The diabatic approach shows that this 'avoided crossing' is not correlated with any crossing between diabatic states in this energy range. In addition, from the adiabatic curves the peak of the radial coupling would have been attributed to this C–D 'avoided crossing', while according to the diabatic results we should assign this peak to the Li(2s)H, Li(3s)H diabatic crossing. The important point here is that following the adiabatic picture one would conclude an accidental interaction between two Rydberg states while from the diabatic picture we see an intrinsic effect due to an over-repulsiveness of the valence diabatic states. We will encounter later the same behaviour for the triplet states. This effect is also probably present for the other alkali hydride molecules and should similarly lead to internal peaks of the radial coupling which may be important in the quenching of the Rydberg states.

The comparison with a very different *ab initio* calculation being satisfactory for these 3 radial couplings, we can pursue an inspection of the new results. The 4–5 radial coupling presents only one peak, rather intense and at short internuclear distances. This peak is not related to a crossing between diabatic states, thus it is due to variations of the electronic coupling. A second peak, associated with the neutral Li(3p)H ionic Li^+H^- diabatic crossing, could be expected around 35 au. Following the analysis of Mendez *et al* [1] this peak and all the peaks at larger distances associated with the neutral-ionic crossings of the Rydberg states higher in energy (i.e. 3d, 4s, 4p, 4d and 4f) are not efficient in producing non-adiabatic transitions. The 5–6, 6–7 and 7–8 couplings present at short internuclear distances peaks which become more and more sharp and intense; they are related to neutral-neutral crossings between diabatic states. For the 5–6 coupling, there is a second broad peak around 6 au, which can be associated with a minimum of the energy difference between the diabatic Li(3d)H and Li(3p)H

curves. It also presents a broad peak around 20 au sensitive to the ionic state energy shift, in a region where the energy curves of these neutral diabatics remain parallel and very close to each other. This peak can be related to a variation of the effective electronic coupling originating from a differential coupling with the ionic state. The peak of the 6-7 coupling around 11 au is more intense and could again be related to a minimum of the energy difference between two diabatic neutral states, here the Li(3d)H and Li(4s)H. The sharp and intense peak of the 7-8 coupling is correlated

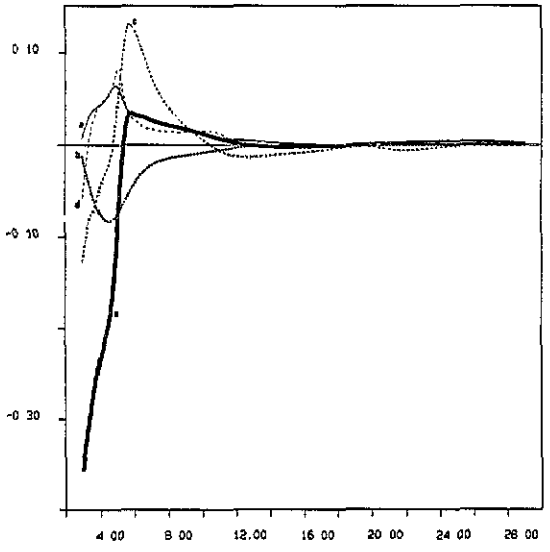


Figure 17. Next-next-neighbour couplings for the $^1\Sigma^+$ adiabatic states a: 1-4; b: 2-5, c: 3-6, d: 4-7 and e: 5-8.

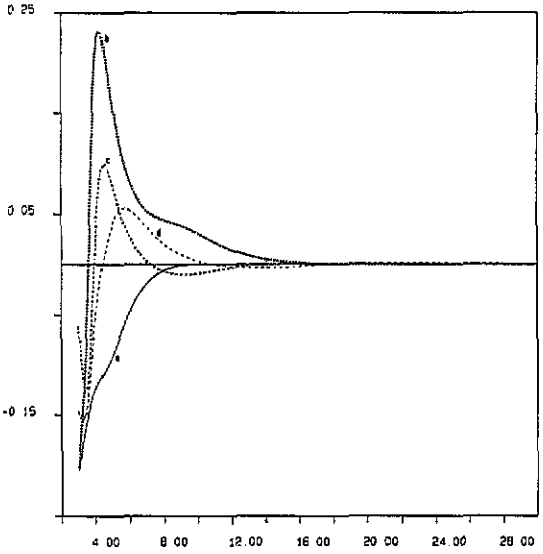


Figure 18. Next-next-neighbour couplings for the $^3\Sigma^+$ adiabatic states a: 9-12, b: 10-13, c: 11-14 and d: 12-15.

with a crossing between the diabatic Li(4s)H and Li(4p)H, however, its location can also be related to the minimum of the energy difference between the seventh and eighth adiabatic states. It should be emphasized that these rather intense peaks of the radial coupling could play an important role in the quenching of the Rydberg states. In slow $Li^* + H$ collisions they probably produce inelastic non-adiabatic transitions which should be accessible to experimental confirmation.

The coupling involving two adiabatic states whose indices differ by two, are less intense except at short interatomic distances for the 4–6 and 5–7 couplings. In this very repulsive region of the curves, as noticed earlier, it can be emphasized that the diabatic state Li(2s)H is more repulsive than the Rydberg diabatic and produces an *intricate cascade of crossings and recrossings, which could hardly be guessed from* the adiabatic curves which behave smoothly. This series of intense peaks of the radial coupling at short distances is its signature.

The same behaviour can be observed for the triplet states. The Li(2s)H diabatic triplet crosses many Rydberg states at short internuclear distances generating a series of peaks in the radial coupling. The presence of a peak of the coupling 9–10 between the first two adiabatic triplets is rather surprising from the adiabatic curves. It can be clearly attributed to the Li(2s)H, Li(2p)H diabatic crossing. The smooth behaviour of the lowest adiabatic curves hides the strong interaction between valence and first Rydberg states, which becomes transparent from the diabatic curves. The sharp and intense peak of the 10–11 coupling is better related to the minimum of the energy difference between the adiabatic triplets labelled c and d in figure 4. The two internal peaks of the 12–13 coupling can be related to the two crossings between the diabatic triplets Li(3d)H and Li(3p)H. This coupling also presents a broad peak in regions where the adiabatic and diabatic curves behave in parallel and are very close in energy. The 13–14 and 14–15 couplings present very intense peaks at short distances. Surprisingly, couplings involving adiabatic triplets whose indices differ by 2 also present such *intense peaks at short internuclear distances. Even the next-next-neighbour couplings* are rather intense at short interatomic distances for the singlets (figure 17) and for the triplets (figure 18). All these peaks result from the intricate crossings of the repulsive part of the diabatic states. Such peaks of the radial coupling could also be important for high-energy collisions where the repulsive part of the curves plays a dominant role. It should be emphasized that most of these peaks in the radial coupling are rather unexpected from the shape of the adiabatic curves.

3. Conclusion

We have computed the radial coupling for the singlets and triplets of Σ^+ symmetry of the LiH molecule dissociating into Li(2s, 2p, 3s, 3p, 3d, 4s, 4p) + H and $Li^+ + H^-$. We used accurate diabatic *ab initio* results and our basic assumption was to neglect an eventual residual coupling. The results for the lowest $^1\Sigma^+$ are very similar in shape, location and intensity to other, more conventional *ab initio* calculations involving very different techniques specifically devoted to such problems. This success for the three first adiabatic singlets shows the validity of our approach and grounds the diabatization method which supports this work. New results are presented for a complete set of singlets and triplets and analysed. The peaks of the radial coupling can easily be assigned to crossings between diabatic states. An interesting new aspect is the presence of numerous neutral-neutral crossings at short internuclear distances leading to intense

peaks of the radial coupling even for adiabatic states whose indices differ by 2 and 3. Most of these crossings are due to an over-repulsiveness of the valence diabatic states as compared to the Rydberg ones, and is probably a common feature in the alkali hydrides. Such peaks could be important in medium and high-energy collisions. At intermediate internuclear distances, in addition to the well known neutral-ionic peaks which play a dominant role in the mutual neutralization process, we observe some intense peaks, especially for the singlets, also related to neutral-neutral interactions. Such couplings can be expected to actively participate in the inelastic quenching of Rydberg states in low-energy $Li^* + H$ collisions. The numerical advantages of this approach are evident. The electronic Hamiltonian in the diabatic basis can be easily interpolated and its derivatives computed since all matrix elements present smooth variations. Then the necessary diagonalization of this small matrix and the eventual differentiations can be easily performed for as many points as necessary. Thus the determination of the radial coupling between the adiabatic states presents no numerical difficulty, even for sharp peaks. The main difficulty is of course the obtention of the electronic Hamiltonian in the diabatic basis. Let us finally emphasize that most of the radial coupling peaks are rather unexpected from the adiabatic picture while the diabatic one illuminates their physical content and origin, and makes the interpretation transparent allowing for generalizations.

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